

PR-AYC-G and Topo-OSM

Open protocol and analysis architecture for state-locked narrative meaning

Working Manuscript v0.4 - Public-launch methods update; exploratory, not confirmatory

Manuscript status and claim boundary

This draft is written in a peer-review style, but it remains a working methods manuscript. PR-AYC-G is presented as an exploratory naturalistic psychophysiology protocol and open-source software architecture. Topo-OSM is presented as a human-scale network-state interpretation layer. Current evidence does not prove consciousness, cellular OSM, microtubular memory, biophotonic memory, clinical efficacy, molecular hidden variables, or literal thermodynamic entropy reduction.

Abstract

PR-AYC-G was developed to test whether accessible narrative meaning produces a measurable, state-locked body-brain trajectory that differs from matched audiovisual stimulation and from the same intact stimulus under analytic task stance. The current protocol compares a phase-scrambled Control, an intact Target narrative, and the same intact narrative under Contextual Override, typically an upper-right number-cue running-sum task. Since the previous working manuscript, the project has been compressed around a clearer primary endpoint: Anchor-Locked MRED (A-MRED), with MRED-Peak and MRED-Resolution as endpoint subtypes. A-MRED asks whether a predeclared scene anchor in the Target branch shows both meaning-recognition evidence and delayed integration/recovery evidence greater than Control and Override after QC, artifact, and stimulus-regressor checks. The software stack now includes MediaPrep/StimulusFingerprint v1.8, PRAYCG2.0 consolidated self-report, the Master Comprehensive Suite v1.5.5, a synchronized visualizer, and an offline interpretive report generator. Recent single-participant pilot analyses suggest three useful archetypes: Contact as the strongest current MRED-Peak case, Field of Dreams as the strongest MRED-Resolution candidate, and Arrival as a recognition-dominant but integration-uncertain run. These are hypothesis-generating self-runs, not confirmatory evidence. The principal contribution is the open, auditable measurement architecture and endpoint hierarchy for future preregistered replication.

1. Current contribution map

Layer	Current permitted claim	Not permitted from current evidence
PR-AYC-G protocol	Open exploratory three-arm design comparing scrambled sensory control, natural Target, and analytic Override.	Does not prove consciousness, souls, love as force, or clinical effect.
A-MRED / MRED-Peak	Candidate primary endpoint for acute Target-specific recognition plus delayed integration at locked anchors.	Does not prove memory formation or mechanism by itself.
MRED-Resolution	Candidate profile for delayed reflective/regulatory recovery and	Does not prove healing, therapy, or structural memory.

	Baseline2 echo.	
Topo-OSM	Human-scale network-state interpretation: meaning may alter measurable state topology.	Does not infer microtubules, biophotons, or molecular hidden-Y from scalp EEG.
NIP/TTI/NUPI	Secondary summaries of immersion proxy, reception/extraction contrast, and narrative update polarity.	Not moral, clinical, neurochemical, or calorimetric scores.
CET-R / StimulusFingerprint	Guardrail for testing whether effects reduce to luminance, cuts, cue rhythm, audio envelope, or motion.	Does not certify meaning or physiology by itself.
Opto-PING Rung 0	Synthetic identifiability under known hidden-loop conditions.	Does not establish human EEG mechanism or OSM biology.

2. Research question

The central empirical question is deliberately narrow:

Core question
When a story is allowed to matter, does the body-brain system show a measurable state trajectory that differs from the same sensory material with meaning disrupted and from the same intact story under analytic extraction?

This question does not require that a single EEG frequency be declared a biomarker of meaning. It requires a multistage test: preserved stimulus identity, stimulus-side fingerprinting, timing validation, artifact control, branch contrasts, and endpoint compression.

3. Experimental architecture

Arm	Stimulus identity	Participant stance	Primary role
Control	Phase-scrambled version of cue-embedded Target, with low-level timing preserved and intelligible meaning disrupted.	Watch normally.	Tests low-level audiovisual timing, luminance, cut rhythm, cue energy, and audio envelope.
Target	Intact meaningful narrative with the same cue-embedded video as Override.	Watch naturally and allow the story to land.	Tests accessible meaning.
Contextual Override	The same intact cue-embedded narrative as Target.	Perform analytic running-sum task while watching.	Tests extraction/task stance against reception.

4. Current software stack

Component	Current role	Key output
MediaPrep + StimulusFingerprint v1.8	Generates Target/Override/Control videos, cue schedules, draft/locked anchors, ALS pulse, and stimulus regressors.	MP4s, cue JSON/CSV, anchor JSON/CSV, qc/stimulus_exogenous_regressor_frame_all_conditions.csv.

PRAYCG2.0 runner	Presents phases, emits StasisMarkers, records media selection hashes, collects consolidated self-report and confounds.	events.csv/json, run config, self-report JSON, confound JSON, override task report.
Master Comprehensive Suite v1.5.5	Computes feature frame, QC tables, A-MRED, MRED-Peak/Resolution, NIP/TI/NUPI/CET/EET/ITP/OCM/RSM outputs.	tables/*.csv, reports/*.md/pdf/docx, offline interpretive report.
MasterSync Visualizer	Creates synchronized public/TSC or lab MP4 review videos with overlays.	MP4 plus features_used/events_used/render report sidecars.
Offline interpretive report generator	Rule-based local plain-English summary of module outputs.	offline_interpretive_report.md/txt/json.

5. Primary endpoint compression: A-MRED

A-MRED is now the recommended primary endpoint for future runs. It converts the overcomplete research workbench into a smaller confirmatory path.

A-MRED rule
A predeclared scene event passes only if Target shows both meaning recognition and delayed integration greater than Control and Override after timing, artifact, confound, and stimulus-rhythm controls. If the anchor is estimated or not frame-verified, the result can be a pilot-positive candidate but not strict confirmation.

The conceptual formula is:

$$\text{A-MRED}_j = 1[\text{MR}_{T,j} > \text{threshold_MR}] * 1[\text{ENC}_{T,j} > \text{threshold_ENC}] * 1[\text{MR}_{T,j} > \text{MR}_{C,j} + \text{delta}] * 1[\text{MR}_{T,j} > \text{MR}_{O,j} + \text{delta}] * 1[\text{ENC}_{T,j} > \text{ENC}_{C,j} + \text{delta}] * 1[\text{ENC}_{T,j} > \text{ENC}_{O,j} + \text{delta}] * 1[\text{QC}_j = \text{PASS}]$$

6. MRED-Peak and MRED-Resolution

Endpoint	Question	Best current archetype	Caution
MRED-Peak	Did the anchor produce an acute meaning-recognition event with delayed/theta/topological integration?	Contact father/avatar region.	Contact anchors were conceptually predeclared but not runner-registered in the original run.
MRED-Resolution	Did the anchor produce delayed reflective/regulatory recovery rather than a sharp peak?	Field of Dreams closure/recovery profile.	Field anchors were runner-loaded but YouTube-estimated and not frame-verified.
Recognition-dominant	Did meaningful recognition appear without strict integration or resolution pass?	Arrival.	Older run lacks PRAYCG2.0 Baseline2 and is ROI-cautioned.

7. Narrative Update Polarity Index (NUPI)

NUPI asks whether an event appears more like accommodative load or resolute recovery. It is not literal thermodynamics, calorimetry, or a clinical healing score. It compares an update-load profile against a regulatory-dividend profile.

Run	Current profile	Interpretive meaning
Contact	High-load-with-recovery	Strong acute semantic intensity with substantial after-state recovery.
Field of Dreams	Resolute-recovery	Lower immediate amplitude but stronger reflective/parasympathetic after-state profile.
Arrival	Polarity unresolved / recognition-dominant	Meaning recognition appears plausible, but older data cannot grade resolution cleanly.

8. Current pilot evidence summary

The current self-run evidence is useful for methods development and hypothesis generation. It is not a confirmatory dataset.

Run	Best use	Current grade
Contact	Gold-plated but not gold-record example; strongest MRED-Peak archetype; future-freeze candidate.	Conceptually predeclared / not runner-registered.
Field of Dreams	Best MRED-Resolution/NUPI recovery candidate; PRAYCG2.0 and blind anchor file loaded.	Runner-registered estimated-anchor pilot; not frame-verified confirmatory.
Arrival	Best recognition-vs-encoding lesson; meaningful/familiar scene reactivation without strict event lock.	Older run; not resolution-gradable.
Her	Confound/legibility/audio caution case; useful for RSM/CVB/SquintProxy development.	Pilot/cautioned.
Snack Attack	Cue-visibility and running-sum burden stress test.	Override microstate development case.

9. Public release posture

The public GitHub/OSF release should be framed as an open-methods package, not as a final empirical proof package. Contact may serve as the demonstration folder labeled gold-plated but not gold-record. The package should include code, templates, documentation, safety/ethics/privacy boundaries, public-safe derived examples, and the no-copyright synthetic demo. It should not include copyrighted movie clips or private raw biosignals by default.

10. Next confirmatory study design

- Use PRAYCG2.0 or later with consolidated self-report and final reflection baseline.
- Generate final videos with MediaPrep/StimulusFingerprint v1.8 or later.

- Frame-verify anchor times on the final rendered Target MP4 before acquisition.
- Load the locked anchor JSON into the runner so markers and hashes are recorded.
- Validate ALS/PT19 full-screen pulse for Control, Target, and Override before running.
- Add EOG/EMG if possible to harden gamma and blink/ocular interpretations.
- Use A-MRED as the primary endpoint; keep NIP/TTI/NUPI secondary; keep KHT/EET/ITP/OCM/RSM exploratory.
- Treat self-report as an independent covariate stream, not proof of internal state.

11. Limitations

- Single-participant self-runs cannot establish generalizability.
- OpenBCI scalp EEG cannot directly measure microtubules, biophotons, dopamine, oxytocin, ATP use, or cellular memory traces.
- Gamma-derived features are artifact-sensitive and must not be interpreted alone.
- Estimated anchors and weak ALS timing downgrade endpoint claims.
- Control audio unintelligibility and visual source-stamp cleanup require manual QC.
- Complexity, blink, EET, and NUPI modules are convergence tools, not proof layers.

12. Conclusion

The mature form of the project is smaller and stronger than the original theoretical ambition. PR-AYC-G is now best described as an open, auditable protocol for testing whether narrative meaning has timing, shape, and physiological consequence. The recommended primary path is A-MRED, with MRED-Peak and MRED-Resolution separating acute meaning shock from delayed closure/recovery. Topo-OSM remains a human-scale interpretation layer; cellular OSM remains quarantined until directly measured.

Appendix A. Source basis for this update

- Working manuscript v0.2/v0.3, including the existing claim-boundary language and software-stack description.
- PRAYCG2.0 runner files from Field of Dreams Run 1, including loaded media paths, anchor file, final reflection baseline, and consolidated reports.
- Master Comprehensive Suite v1.5.5 design: A-MRED, MRED-Peak/Resolution, offline interpreter, and module-tier compression.
- Contact, Field, Arrival, NUPI, and MRED-Peak/Resolution analysis reports generated in this workflow.
- MediaPrep reports documenting Target/Override identity, phase-scrambled Control, ALS timing square, cue schedules, and QC boundaries.